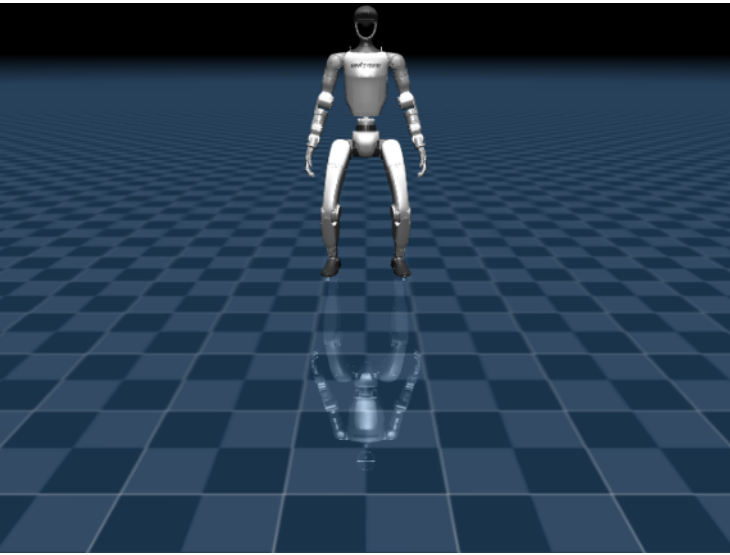
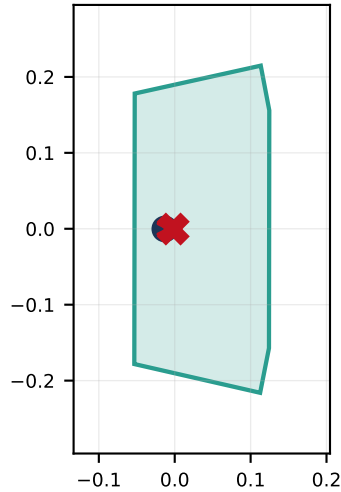


Physical challenges in Kalaripayattu execution (real MuJoCo states; physics from the validated Pinocchio runtime)

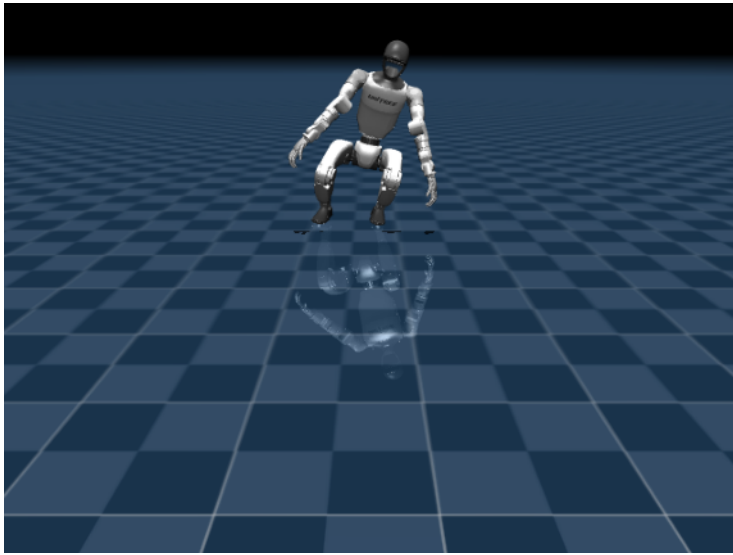
A deep stance (double support)



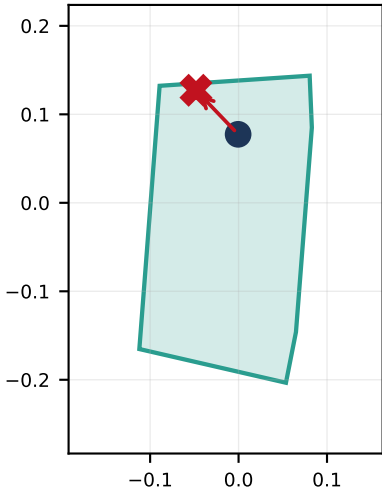
$R_{cp} = +0.051 \text{ m}$  mode: nominal



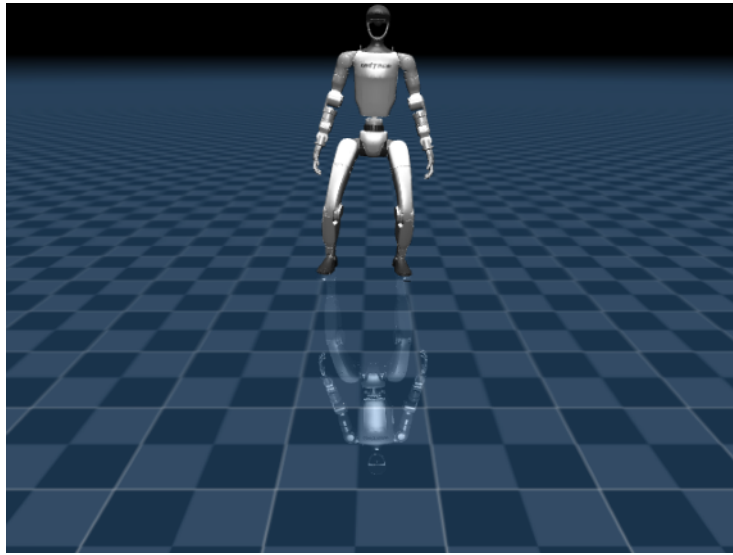
B 120 N push: CP escaping support



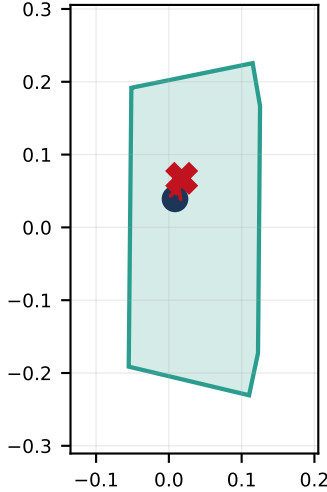
$R_{cp} = +0.008 \text{ m}$  mode: fall



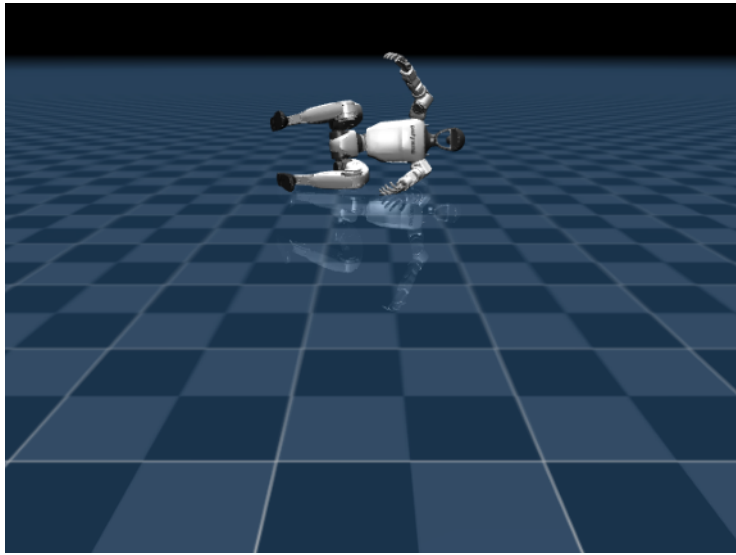
C 100 N push: CP held inside



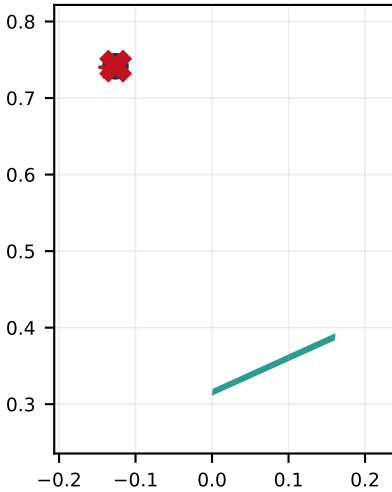
$R_{cp} = +0.070 \text{ m}$  mode: nominal



D post-fall protective state



$R_{cp} = -0.439 \text{ m}$  mode: recovery



support polygon   CoM (xy)   capture point  $\xi$